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Vehicle body for vehicle

Abstract

A vehicle body for a vehicle, includes a first body, a second body configured to be fitted with the first body, an insertion portion provided on one of the first body and the second body and including an insertion space formed therein, and a fixing portion provided on the other of the first body and the second body and configured to be inserted into the insertion space of the insertion portion, the fixing portion including a magnetic module provided at an end portion of the fixing portion and configured to fix the fitted first and second bodies by use of a magnetic force generated by a change in magnetic circuit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
(1) The present application claims priority to Korean Patent Application No. 10-2022-0099407, filed on Aug. 9, 2022, the entire contents of which is incorporated herein for all purposes by this reference.
BACKGROUND OF THE PRESENT DISCLOSURE
Field of the Present Disclosure
(2) The present disclosure relates to a vehicle body for a vehicle, and more particularly, to a technology for coupling a plurality of bodies that forms a vehicle body.
Description of Related Art
(3) Recently, new mobility visions with new concepts for implementing human-oriented dynamic future cities have been introduced to vehicle industries. One of the future mobility solutions is a purpose-built vehicle (PBV) as a purpose-based mobility vehicle.
(4) The PBV is an environmental-friendly movement solution for providing necessary customized services to an occupant while moving to the destination on the ground. The PBV may set an optimum route for each situation and perform platooning by use of electric vehicle-based, artificial

intelligence.
(5) The PBV vehicle includes a life module in which a passenger or a luggage is loaded, and a driving module in which a drive device for a vehicle is disposed. The life module and the driving module are fastened to each other to form a vehicle body.
(6) A vehicle body for a PBV vehicle in the related art has a problem in that it is difficult to couple a plurality of bodies by mechanical coupling, such as welding or bolting, and to decouple the plurality of bodies, which makes it difficult to manage and repair the vehicle.
(7) The information included in this Background of the present disclosure is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.
BRIEF SUMMARY
(8) Various aspects of the present disclosure are directed to providing a vehicle body for a vehicle, which is implemented as a first body and a second body are fitted with each other so that the vehicle body is conveniently disassembled or assembled because the first body and the second body are fixed by a magnetic module in a state in which the first body and the second body are fitted with each other.
(9) Various aspects of the present disclosure are directed to providing a vehicle body for a vehicle, the vehicle body including: a first body; a second body configured to be fitted with the first body; an insertion portion provided on one of the first body and the second body and including an insertion space formed therein; and a fixing portion provided on the other of the first body and the second body and configured to be inserted into the insertion space of the insertion portion, the fixing portion including a magnetic module provided at an end portion of the fixing portion and configured to fix the fitted first and second bodies by use of a magnetic force generated by a change in magnetic circuit.
(10) The fixing portion and the insertion portion may be disposed to correspond to each other at edge portions of the first body and the second body that are coupled to each other.
(11) The magnetic module may be disposed so that the magnetic force is directed in a direction intersecting a direction in which the fixing portion is inserted.
(12) The insertion portion may include a protruding portion protruding from an internal surface of the insertion portion so that the magnetic module is caught by the protruding portion when the magnetic module is fixed to the insertion portion.
(13) The protruding portion may be formed on an internal surface of the vehicle by press forming.
(14) The fixing portion may further include: a hinge unit extending in a direction intersecting the direction in which the fixing portion is inserted; and a link unit rotatably coupled to an end portion of the hinge unit and extending in the direction in which the fixing portion is inserted, and the magnetic module may be disposed at one end portion of the link unit in a direction intersecting the direction in which the link unit extends.
(15) The fixing portion may further include an elastic member connected to the link unit and configured to apply an elastic force in a direction opposite to a direction of the magnetic force of the magnetic module.
(16) The hinge unit may be provided as a pair of hinge units, the link unit may be provided as a pair of link units, the magnetic module may be provided as a pair of magnetic modules, and the pair of hinge units, the pair of link units, and the pair of magnetic modules may be disposed in the direction intersecting the direction in which the fixing portion is inserted.
(17) The fixing portion may further include an elastic member connected to the link unit and configured to apply an elastic force in a direction opposite to a direction of the magnetic force of magnetic module, and the elastic member may connect the pair of link units and apply the elastic force in the direction opposite to the direction of the magnetic force of the magnetic module.
(18) The fixing portion may further include a base unit coupled to the other of the first and second

bodies, and the hinge unit may extend from the base unit.
(19) The fixing portion may further include a stepped portion formed to be stepped and configured to come into contact with an end portion of the insertion portion, the stepped portion being configured to restrict the insertion of the fixing portion.
(20) The vehicle body may further include a connector electrically connected to the magnetic module and configured to operate the magnetic module by applying electric current to the magnetic module.
(21) The magnetic module may include: a module main body configured to come into contact with the insertion portion; a coil wound around the module main body; a stationary magnetic element coupled to the module main body; and a rotary magnetic element rotatably coupled to the module main body, the connector may be connected to the coil, a direction of a magnetic field formed by the coil may be changed depending on a flow direction of the electric current of the connector, and the rotary magnetic element may rotate so that the magnetic circuit is changed.
(22) The first body may have a space in which a passenger is accommodated or a luggage is loaded, and the second body may include a driving unit of driving a vehicle and be provided as a pair of second bodies which is coupled to the first body in a forward/rearward direction thereof.
(23) According to the vehicle body for a vehicle according to an exemplary embodiment of the present disclosure, to couple the first and second bodies, the insertion portion is provided on one of the first and second bodies, the fixing portion including the magnetic modules is provided on the other of the first and second bodies, and the fixing portion is inserted into the insertion portion and thus fitted with the insertion portion. Furthermore, the magnetic modules operate in the state in which the fixing portion is fitted with the insertion portion, and the fixing portion and the insertion portion are fixed by the magnetic force. Therefore, it is possible to easily fasten the first and second bodies.
(24) Furthermore, when the magnetic module is controlled to eliminate the magnetic force, the first body and the second body may be easily unfastened so that when the vehicle body is damaged or needs to be replaced, the first body and the second body are uncoupled, and then the vehicle body is repaired or replaced. Therefore, it is possible to improve maintainability of the vehicle body.
(25) Furthermore, the first body has the space in which a passenger is accommodated or a luggage is loaded, and the second body includes the driving unit of driving the vehicle. Furthermore, the second body is provided as a pair of second bodies which may be coupled to the first body in the forward/rearward direction thereof. Therefore, the first body and the second body may be conveniently coupled and decoupled, and the first body and the second body may be freely replaced in accordance with the customer's requirement, improving the marketability of the vehicle.
(26) The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a vehicle body for a vehicle according to various exemplary embodiments of the present disclosure.
(2) FIG. **2** is a cross-sectional view exemplarily illustrating a state in which an insertion portion and a fixing portion included in the vehicle body for a vehicle according to the exemplary embodiment of the present disclosure are not coupled.
(3) FIG. **3** is a perspective view exemplarily illustrating a state in which the insertion portion and the fixing portion included in the vehicle body for a vehicle according to the exemplary

embodiment of the present disclosure are coupled.
(4) FIG. **4**A and FIG. **4**B are cross-sectional views exemplarily illustrating an operation of a magnetic module included in the vehicle body for a vehicle according to the exemplary embodiment of the present disclosure.
(5) It may be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as included herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined in part by the particularly intended application and use environment.
(6) In the figures, reference numbers refer to the same or equivalent parts of the present disclosure throughout the several figures of the drawing.
DETAILED DESCRIPTION
(7) Reference will now be made in detail to various embodiments of the present disclosure(s), examples of which are illustrated in the accompanying drawings and described below. While the present disclosure(s) will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that the present description is not intended to limit the present disclosure(s) to those exemplary embodiments of the present disclosure. On the other hand, the present disclosure(s) is/are intended to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims.
(8) Hereinafter, embodiments included in the present specification will be described in detail with reference to the accompanying drawings. The same or similar constituent elements are assigned with the same reference numerals regardless of reference numerals, and the repetitive description thereof will be omitted.
(9) The suffixes ‘module’, ‘unit’, ‘part’, and ‘portion’ used to describe constituent elements in the following description are used together or interchangeably to facilitate the description, but the suffixes themselves do not have distinguishable meanings or functions.
(10) In the description disclosed in the present specification, the specific descriptions of publicly known related technologies will be omitted when it is determined that the specific descriptions may obscure the subject matter of the exemplary embodiments disclosed in the present specification. Furthermore, it should be understood that the accompanying drawings are provided only to allow those skilled in the art to easily understand the exemplary embodiments disclosed in the present specification, and the technical spirit disclosed in the present specification is not limited by the accompanying drawings, and includes all alterations, equivalents, and alternatives that are included in the spirit and the technical scope of the present disclosure.
(11) The terms including ordinal numbers such as “first,” “second,” and the like may be used to describe various constituent elements, but the constituent elements are not limited by the terms. These terms are used only to distinguish one constituent element from another constituent element.
(12) When one constituent element is referred to as being “coupled” or “connected” to another constituent element, it should be understood that one constituent element may be coupled or directly connected to another constituent element, and an intervening constituent element can also be present between the constituent elements. When one constituent element is referred to as being “directly coupled to” or “directly connected to” another constituent element, it may be understood that no intervening constituent element is present between the constituent elements.
(13) Singular expressions include plural expressions unless clearly referred to as different meanings in the context.
(14) In the present specification, it should be understood the terms “comprises,” “comprising,” “includes,” “including,” “containing,” “has,” “having” or other variations thereof are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, components,

or combinations thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.
(15) FIG. **1** is a perspective view of a vehicle body for a vehicle according to various exemplary embodiments of the present disclosure, FIG. **2** is a cross-sectional view exemplarily illustrating a state in which an insertion portion and a fixing portion included in the vehicle body for a vehicle according to the exemplary embodiment of the present disclosure are not coupled, FIG. **3** is a perspective view exemplarily illustrating a state in which the insertion portion and the fixing portion included in the vehicle body for a vehicle according to the exemplary embodiment of the present disclosure are coupled, and FIG. **4**A and FIG. **4**B are cross-sectional views exemplarily illustrating an operation of a magnetic module included in the vehicle body for a vehicle according to the exemplary embodiment of the present disclosure.
(16) An exemplary embodiment of the vehicle body for a vehicle according to an exemplary embodiment of the present disclosure will be described with reference to FIG. **1**, FIG. **2**, FIG. **3**, and FIG. **4**A and FIG. **4**B.
(17) In the case of a vehicle body generally, a plurality of bodies may be fixed by welding in a state in which the plurality of bodies is coupled, or a plurality of bodies may be fastened by mechanical coupling such as bolting.
(18) This structure makes it difficult to repair the vehicle body and makes it impossible to replace the vehicle body when the vehicle body is broken or a portion of the vehicle body is damaged due to a vehicle accident.
(19) Various embodiments of the present disclosure relates to a vehicle body coupling structure for solving the above-mentioned problem, and various aspects of the present disclosure are directed to providing a configuration in which magnetic modules **310** are applied to fixing portions and coupled to each other so that bodies forming a vehicle body are conveniently coupled or decoupled.
(20) A vehicle body for a vehicle according to various exemplary embodiments of the present disclosure may include a first body **100**; second bodies **200** configured to be fitted with the first body **100**; insertion portions **400** provided on any one of the first body and the second body **100** and **200** and each including an insertion space formed therein; and fixing portions **300** provided on the other of the first body and the second body **100** and **200** and each configured to be inserted into the insertion space of each of the insertion portions **400**, the fixing portions **300** each including magnetic modules **310** provided at an end portion of the fixing portion **300** and configured to fix the fitted first and second bodies **100** and **200** by use of a magnetic force generated by a change in magnetic circuit.
(21) As illustrated in FIG. **1**, FIG. **2**, and FIG. **3**, the first body and the second body **100** and **200** are coupled by being fitted with each other, forming the vehicle body for a vehicle. In the instant case, any one of the first body and the second body **100** and **200** has the insertion portion **400** including the insertion space into which an external object may be inserted. The other of the first body and the second body **100** and **200** has the coupling device configured to be inserted into the insertion space. Therefore, the first body and the second body **100** and **200** are coupled to each other as the coupling device and the insertion portion **400** are fitted with each other so that the vehicle body may be assembled.
(22) The insertion portion **400** may be provided on any one of the first body and the second body **100** and **200**. The fixing portion **300** may be provided on the body that does not have the insertion portion **400**. The fixing portion **300** may be provided at a position corresponding to the insertion portion **400** and fitted with the insertion portion **400**.
(23) In the instant case, the magnetic modules **310** may be provided on the coupling device and generate the magnetic force outward by changing the magnetic circuit. The magnetic modules **310** operate in the state in which the coupling device is fitted with the insertion portion **400** so that the fixing portion **300** and the insertion portion **400** may be fixed to each other in the insertion space.
(24) Therefore, the first body and the second body **100** and **200** may be strongly fixed to each other

by the magnetic modules **310** in the state in which the first body and the second body **100** and **200** are fitted with each other. When the first body and the second body **100** and **200** coupled to each other need to be decoupled, as necessary, to repair or replace the first body **100** or the second body **200**, the magnetic modules **310** are controlled to easily unfix and separate the first body and the second body **100** and **200** so that the first body **100** or the second body **200** may be conveniently repaired or replaced.

(25) As illustrated in FIG. **1**, in the exemplary embodiment of the present disclosure, the first body **100** has a space in which a passenger is accommodated or a luggage is loaded, and the second body **200** includes a driving unit of driving the vehicle. The second body **200** may be provided as a pair of second bodies which may be coupled to the first body **100** in a forward/rearward direction thereof.

(26) In the exemplary embodiment of the present disclosure, the first body **100** may be a life module including a seating space in which a passenger may be accommodated or a loading space in which luggage may be loaded. The second body **200** may be configured as a driving module including the driving unit of the vehicle. The second bodies **200** may be coupled to the first body **100** in the forward/rearward direction and form front and rear driving units of the vehicle.

(27) Therefore, the first body and the second body **100** and **200** are coupled and fixed by the magnetic modules **310**, forming the vehicle body for a vehicle. Because the fixing portion **300** and the insertion portion **400** are easily decoupled, the first body **100**, which is the life module, or the second body **200**, which is the driving module, is easily replaced in accordance with a customer's requirement so that marketability of the vehicle may be improved. Furthermore, because the second body **200**, which is the driving module which is more frequently broken down than the life module, may be easily replaced, maintainability of the vehicle may be improved.

(28) The fixing portion **300** and the insertion portion **400** may be provided to correspond to each other at the edge portions of the first body and the second body **200** and **100** that are coupled to each other.

(29) As illustrated in FIG. **1**, in the exemplary embodiment of the present disclosure, the insertion portion **400** may be provided on the first body **100**, and the fixing portion **300** may be provided on the second body **200** and disposed to correspond to the position of the insertion portion **400**.

(30) In the instant case, the insertion portion **400** may be provided as a plurality of insertion portions, the fixing portion is provided as a plurality of fixing portions, and the plurality of insertion portions and the plurality of fixing portions are respectively provided to correspond to one another and disposed at the edge portions of the coupling surfaces of the first body and the second body **100** and **200** so that the first body and the second body **100** and **200** may be securely coupled.

(31) The magnetic modules **310** may be disposed so that the magnetic force is directed in a direction intersecting a direction in which the fixing portion **300** is inserted.

(32) As illustrated in FIG. **2** and FIG. **3**, the magnetic modules **310** may be disposed in a direction intersecting the direction in which the fixing portion **300** is inserted. The magnetic modules **310** may be disposed so that the magnetic force is generated in a direction perpendicular to the direction in which the fixing portion **300** is inserted.

(33) Therefore, after the fixing portion **300** is inserted into the insertion space of the insertion portion **400**, the magnetic modules **310** are controlled to generate the magnetic force so that the magnetic modules **310** may come into contact with an internal surface of the insertion portion **400** and be immediately coupled by the magnetic force.

(34) In the instant case, the insertion portion **400** may be made of metal, which is a diamagnetic element, so that the insertion portion **400** is fixed by the magnetic force of the magnetic module **310**.

(35) A protruding portion **410** may protrude from the internal surface of the insertion portion **400** so that the magnetic module **310** is caught by the protruding portion **410** when the magnetic module **310** is fixed.

(36) As illustrated in FIG. **3**, the protruding portion **410** protrudes from the internal surface of the insertion portion **400**, and the protruding portion **410** may be disposed outwardly from a position at which the magnetic module **310** is fixed when the fixing portion **300** is inserted into the insertion portion **400**.
(37) There may occur a problem in that the magnetic module **310** slides on the internal surface of the insertion portion **400** and the insertion portion **400** and the fixing portion **300** are unfastened even though the magnetic module **310** is fixed to the internal surface of the insertion portion **400** by the magnetic force.
(38) When the magnetic module **310** slides along the internal surface of the insertion portion **400** as described above, the magnetic module **310** is caught by the protruding portion **410** so that the fixing portion **300** may be simultaneously and doubly fixed magnetically and mechanically.
(39) The protruding portion **410** may be formed on an internal surface of the vehicle by press forming.
(40) As illustrated in FIG. **3**, the protruding portion **410** protruding inwardly from the internal surface of the insertion portion **400** may be formed by press forming. The protruding portion **410** may be formed on an internal side surface of the vehicle by press forming.
(41) Therefore, an external surface of the vehicle may define an external appearance of the vehicle without including the protruding portion **410**. A portion at the internal side of the vehicle, which is recessed when the protruding portion **410** is formed by press forming, may be covered by an internal material provided in the interior of the vehicle.
(42) The fixing portion **300** may further include: hinge units **320** extending in a direction intersecting the direction in which the fixing portion **300** is inserted; and link units **330** rotatably coupled to end portions of the hinge units **320** and extending in the direction in which the fixing portion **300** is inserted. The magnetic module **310** may be disposed at one end portion of the link unit **330**. The magnetic modules **310** may be disposed in the direction intersecting the direction in which the link units **330** extend.
(43) As illustrated in FIG. **2** and FIG. **3**, the hinge unit **320** may extend in the direction perpendicularly intersecting the direction in which the fixing portion **300** is inserted. The link unit **330** may be rotatably coupled to the end portion of the hinge unit **320** and extend in the direction in which the fixing portion **300** is inserted. The magnetic module **310** may be provided at one end portion of the link unit **330**. In the instant case, the magnetic module **310** may be provided to be directed toward the internal surface of the insertion portion **400**.
(44) The link unit **330** rotates about a point at which the link unit **330** is connected to the hinge unit **320**. Therefore, the magnetic module **310** is spaced from the internal surface of the insertion portion **400** at the time of inserting the fixing portion **300** into the insertion portion **400**. When the fixing portion **300** is completely inserted, the magnetic module **310** operates so that the link unit **330** is rotated by the magnetic force, and the magnetic module **310** may be fixed to the internal surface of the insertion portion **400**.
(45) Therefore, the magnetic module **310** is not in contact with the internal surface of the insertion portion **400** while the fixing portion **300** is inserted into the insertion portion **400**. Therefore, because the magnetic module **310** is not in contact with the internal surface of the insertion portion **400** or the protruding portion **410**, the breakdown of or damage to the magnetic module **310** may be prevented, and the fixing portion **300** may be easily inserted.
(46) The fixing portion **300** may further include an elastic member **340** connected to the link unit **330** and configured to apply an elastic force in a direction opposite to the direction of the magnetic force of the magnetic module **310**.
(47) The elastic member **340** rotates the link unit **330** by applying the elastic force to the link unit **330** so that the magnetic module **310** is maintained to be spaced from the internal surface of the insertion portion **400**. When the magnetic module **310** is controlled to change the magnetic circuit, the magnetic force is generated, and the magnetic force of the magnetic module **310** is higher than

the elastic force of the elastic member **340** so that the link unit **330** may rotate as the elastic member **340** is extended, and the magnetic module **310** may be fixed to the internal surface of the insertion portion **400**.
(48) Therefore, when the magnetic module **310** is unfixed as the magnetic circuit is controlled, the magnetic module **310** may be moved away from the internal surface of the insertion portion **400** by the elasticity of the elastic member **340** so that the convenience may be improved at the time of separating the fixing portion **300** and the insertion portion **400**.
(49) The hinge unit **320** may be provided as a pair of hinge units, the link unit **330** may be provided as a pair of link units, the magnetic module **310** may be provided as a pair of magnetic modules, and the pair of hinge units, the pair of link units, and the pair of magnetic modules may be disposed in the direction intersecting the direction in which the fixing portion **300** is inserted.
(50) As illustrated in FIG. **2** and FIG. **3**, the pair of hinge units **320**, the pair of link units **330**, and the pair of magnetic modules **310**, which are included in the fixing portion **300**, may be disposed symmetrically in the direction intersecting the direction in which the fixing portion **300** is inserted into the insertion portion **400**. When the magnetic module **310** is controlled so that the magnetic circuit is changed, the pair of magnetic modules **310** is fixed to the internal surface of the insertion portion **400** so that the fixing force of the fixing portion **300** may increase.
(51) The fixing portion **300** may further include the elastic member **340** connected to the link units **330** and configured to apply the elastic force in the direction opposite to the direction of the magnetic force of the magnetic modules **310**. The elastic member **340** may connect the pair of link units **330** and apply the elastic force in the direction opposite to the direction of the magnetic force of the magnetic module **310**.
(52) The elastic member **340** connects the pair of link units **330** and applies the elastic force so that the pair of magnetic modules **310** moves away from the internal surface of the insertion portion **400**. When the magnetic modules **310** operate, the magnetic force becomes higher than the elastic force of the elastic member **340** so that the elastic member **340** may be extended, and the magnetic modules **310** may be fixed to the internal surface of the insertion portion **400**.
(53) The fixing portion **300** may further include base units **370** coupled to the other side body, and the hinge units **320** extend from the base units **370**.
(54) As illustrated in FIG. **2**, FIG. **3** and FIG. **4**A and FIG. **4**B, the hinge unit **320** extends from the base unit **370**, and the base unit **370** may be mechanically coupled to the other side body.
(55) Therefore, the magnetic module **310** may be repaired by separating the base unit **370** from the other side body. Alternatively, when the first body and the second body **100** and **200** cannot be separated due to a breakdown in the state in which the magnetic modules **310** are fixed, the base unit **370** may be separated to repair the magnetic module **310** so that maintainability may be improved.
(56) Furthermore, because the magnetic module **310**, the hinge unit **320**, the link unit **330**, and the base unit **370** are manufactured by modularized into a single module, the process of manufacturing the vehicle body may be simplified.
(57) The fixing portion **300** may further include stepped portions **350** formed to be stepped and configured to come into contact with the end portion of the insertion portion **400** to restrict the insertion of the fixing portion **300**.
(58) As illustrated in FIG. **2**, FIG. **3** and FIG. **4**A and FIG. **4**B, the stepped portion **350** formed to be stepped to restrict the insertion of the insertion portion **400** is provided at the end portion of the fixing portion **300**, and the stepped portion **350** may restrict the insertion of the fixing portion **300**. The external surfaces of the first body and the second body **100** and **200** extend because of stepped shapes of the stepped portions **350**, defining an external aesthetic appearance.
(59) The fixing portion may further include a connector **360** configured to operate the magnetic modules **310** by applying electric current to the magnetic modules **310**.
(60) The connector **360** may be connected to a control unit and a power source unit of the vehicle.

The connector **360** may generate the magnetic force by applying the electric current to the magnetic module **310** and changing the magnetic circuit of the magnetic module **310**.
(61) The connector **360** may be provided on the other side body including the fixing portion **300** and connected to the magnetic module **310** by wiring.
(62) The magnetic module **310** may include: a module main body **311** configured to come into contact with the insertion portion **400**; a coil **312** wound around the module main body **311**; a stationary magnetic element **314** coupled to the module main body **311**; and a rotary magnetic element **313** rotatably coupled to the module main body **311**. The connector **360** is connected to the coil **312**. A direction of a magnetic field formed by the coil **312** is changed depending on a flow direction of the electric current of the connector **360**, and the rotary magnetic element **313** rotates so that the magnetic circuit may be changed.
(63) As illustrated in FIG. **4**A and FIG. **4**B, the magnetic module **310** includes: the module main body **311** configured to come into contact with the internal surface of the insertion portion **400**; the stationary magnetic element **314** fixedly coupled to the module main body **311**; the rotary magnetic element **313** configured to be rotatable; and the coil **312** connected to the connector **360** and configured to be wound around the module main body **311**. When the electric current flows through the coil **312**, the direction of the magnetic field is changed depending on the flow of electric current so that the rotary magnetic element may rotate.
(64) In FIG. **4**B, when the rotary magnetic element **313** rotates so that the magnetic pole of the stationary magnetic element **314** and the magnetic pole adjacent to the magnetic pole of the stationary magnetic element **314** have a same polarity, the magnetic circuit is formed outside the module main body, and the magnetic force is generated so that the module main body may be coupled to an external component by the magnetic force. Furthermore, in FIG. **4**A, when the rotary magnetic element **313** rotates so that the magnetic pole of the stationary magnetic element **314** and the magnetic pole adjacent to the magnetic pole of the stationary magnetic element **314** have different polarities, the magnetic circuit is positioned only in the module main body **311** so that the module main body **311** coupled by the magnetic force may be uncoupled.
(65) In the instant case, the electric current, which may rotate the rotary magnetic element **313**, may be applied to the coil **312**, and the supply of electric current is cut off after the rotary magnetic element **313** is rotated so that the electric power may be effectively used. Therefore, the above-mentioned configuration is more efficient than an electromagnet in the related art.
(66) Furthermore, the term related to a control device such as “controller”, “control apparatus”, “control unit”, “control device”, “control module”, or “server”, etc refers to a hardware device including a memory and a processor configured to execute one or more steps interpreted as an algorithm structure. The memory stores algorithm steps, and the processor executes the algorithm steps to perform one or more processes of a method in accordance with various exemplary embodiments of the present disclosure. The control device according to exemplary embodiments of the present disclosure may be implemented through a nonvolatile memory configured to store algorithms for controlling operation of various components of a vehicle or data about software commands for executing the algorithms, and a processor configured to perform operation to be described above using the data stored in the memory. The memory and the processor may be individual chips. Alternatively, the memory and the processor may be integrated in a single chip. The processor may be implemented as one or more processors. The processor may include various logic circuits and operation circuits, may process data according to a program provided from the memory, and may generate a control signal according to the processing result.
(67) The control device may be at least one microprocessor operated by a predetermined program which may include a series of commands for carrying out the method included in the aforementioned various exemplary embodiments of the present disclosure.
(68) The aforementioned invention can also be embodied as computer readable codes on a computer readable recording medium. The computer readable recording medium is any data

storage device that can store data which may be thereafter read by a computer system and store and execute program instructions which may be thereafter read by a computer system. Examples of the computer readable recording medium include Hard Disk Drive (HDD), solid state disk (SSD), silicon disk drive (SDD), read-only memory (ROM), random-access memory (RAM), CD-ROMs, magnetic tapes, floppy discs, optical data storage devices, etc and implementation as carrier waves (e.g., transmission over the Internet). Examples of the program instruction include machine language code such as those generated by a compiler, as well as high-level language code which may be executed by a computer using an interpreter or the like.
(69) In various exemplary embodiments of the present disclosure, each operation described above may be performed by a control device, and the control device may be configured by a plurality of control devices, or an integrated single control device.
(70) In various exemplary embodiments of the present disclosure, the scope of the present disclosure includes software or machine-executable commands (e.g., an operating system, an application, firmware, a program, etc.) for facilitating operations according to the methods of various embodiments to be executed on an apparatus or a computer, a non-transitory computer-readable medium including such software or commands stored thereon and executable on the apparatus or the computer.
(71) In various exemplary embodiments of the present disclosure, the control device may be implemented in a form of hardware or software, or may be implemented in a combination of hardware and software.
(72) Furthermore, the terms such as “unit”, “module”, etc. included in the specification mean units for processing at least one function or operation, which may be implemented by hardware, software, or a combination thereof.
(73) For convenience in explanation and accurate definition in the appended claims, the terms “upper”, “lower”, “inner”, “outer”, “up”, “down”, “upwards”, “downwards”, “front”, “rear”, “back”, “inside”, “outside”, “inwardly”, “outwardly”, “interior”, “exterior”, “internal”, “external”, “forwards”, and “backwards” are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood that the term “connect” or its derivatives refer both to direct and indirect connection.
(74) The foregoing descriptions of specific exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described in order to explain certain principles of the invention and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

Claims

1. A vehicle body for a vehicle, the vehicle body comprising: a first body; a second body configured to be fitted with the first body; an insertion portion provided on one of the first body and the second body and including an insertion space formed therein; and a fixing portion provided on another of the first body and the second body and configured to be inserted into the insertion space of the insertion portion, the fixing portion including a magnetic module provided at an end portion of the fixing portion and configured to fix the fitted first and second bodies by use of a magnetic force generated by a change in magnetic circuit, wherein the insertion portion includes a protruding portion protruding from an internal surface of the insertion portion so that the magnetic module is caught by the protruding portion when the magnetic module is fixed to the insertion

portion.

2. The vehicle body of claim 1, wherein the fixing portion and the insertion portion are disposed to correspond to each other at edge portions of the first body and the second body that are coupled to each other.

3. The vehicle body of claim 1, wherein the magnetic module is disposed so that the magnetic force is directed in a direction intersecting a direction in which the fixing portion is inserted.

4. The vehicle body of claim 1, wherein the protruding portion is formed on an internal surface of the vehicle by press forming.

5. The vehicle body of claim 3, wherein the fixing portion further includes: a hinge unit extending in a direction intersecting the direction in which the fixing portion is inserted; and a link unit rotatably coupled to an end portion of the hinge unit and extending in the direction in which the fixing portion is inserted, and wherein the magnetic module is disposed at one end portion of the link unit in the direction intersecting the direction in which the link unit extends.

6. The vehicle body of claim 5, wherein the fixing portion further includes an elastic member connected to the link unit and configured to apply an elastic force in a direction opposite in a direction of the magnetic force of the magnetic module.

7. The vehicle body of claim 5, wherein the hinge unit is provided as a pair of hinge units facing each other, the link unit is provided as a pair of link units facing each other, the magnetic module is provided as a pair of magnetic modules facing each other, and the pair of hinge units, the pair of link units, and the pair of magnetic modules are disposed in the direction intersecting the direction in which the fixing portion is inserted.

8. The vehicle body of claim 7, wherein the fixing portion further includes: an elastic member connected to the link unit and configured to apply an elastic force in a direction opposite in a direction of the magnetic force of magnetic module, and wherein the elastic member connects the pair of link units and applies the elastic force in the direction opposite to the direction of the magnetic force of the magnetic module.

9. The vehicle body of claim 5, wherein the fixing portion further includes: a base unit coupled to another one of the first and second bodies, and the hinge unit extends from the base unit.

10. The vehicle body of claim 1, wherein the fixing portion further includes: a stepped portion formed to be stepped and configured to come into contact with an end portion of the insertion portion, the stepped portion being configured to restrict the insertion of the fixing portion.

11. The vehicle body of claim 1, further including: a connector electrically connected to the magnetic module and configured to operate the magnetic module by applying electric current to the magnetic module.

12. The vehicle body of claim 11, wherein the magnetic module includes: a module main body configured to come into contact with the insertion portion; a coil wound around the module main body; a stationary magnetic element coupled to the module main body; and a rotary magnetic element rotatably coupled to the module main body, wherein the connector is connected to the coil, and wherein a direction of a magnetic field formed by the coil is changed depending on a flow direction of the electric current of the connector, and the rotary magnetic element rotates so that the magnetic circuit is changed.

13. The vehicle body of claim 1, wherein the first body has a space in which a passenger is accommodated or a luggage is loaded, and the second body includes a driving unit of driving the vehicle and is provided as a pair of second bodies which is coupled to the first body in forward and rearward direction thereof.

14. A vehicle body for a vehicle, the vehicle body comprising: a first body; a second body configured to be fitted with the first body; an insertion portion provided on one of the first body and the second body and including an insertion space formed therein; and a fixing portion provided on another of the first body and the second body and configured to be inserted into the insertion space of the insertion portion, the fixing portion including a magnetic module provided at an end

portion of the fixing portion and configured to fix the fitted first and second bodies by use of a magnetic force generated by a change in magnetic circuit, wherein the magnetic module is disposed so that the magnetic force is directed in a direction intersecting a direction in which the fixing portion is inserted, wherein the fixing portion further includes: a hinge unit extending in a direction intersecting the direction in which the fixing portion is inserted; and a link unit rotatably coupled to an end portion of the hinge unit and extending in the direction in which the fixing portion is inserted, and wherein the magnetic module is disposed at one end portion of the link unit in the direction intersecting the direction in which the link unit extends.

15. A vehicle body for a vehicle, the vehicle body comprising: a first body; a second body configured to be fitted with the first body; an insertion portion provided on one of the first body and the second body and including an insertion space formed therein; a fixing portion provided on another of the first body and the second body and configured to be inserted into the insertion space of the insertion portion, the fixing portion including a magnetic module provided at an end portion of the fixing portion and configured to fix the fitted first and second bodies by use of a magnetic force generated by a change in magnetic circuit; and a connector electrically connected to the magnetic module and configured to operate the magnetic module by applying electric current to the magnetic module, wherein the magnetic module includes: a module main body configured to come into contact with the insertion portion; a coil wound around the module main body; a stationary magnetic element coupled to the module main body; and a rotary magnetic element rotatably coupled to the module main body, wherein the connector is connected to the coil, and wherein a direction of a magnetic field formed by the coil is changed depending on a flow direction of the electric current of the connector, and the rotary magnetic element rotates so that the magnetic circuit is changed.